

Impact of **Quality Control Tools**

Case: Wossi Garment Factory

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Abstract:

- **Overview:**

The purpose of this paper is to investigate quality control and supply chain management practice in the case company (Wossi Garment Factory) and apply the statistical quality control tools to solve the process and product quality problems in shirt manufacturing and increase the competitiveness of the company.

- **How the case study was conducted:**

The case study was done using both qualitative and quantitative approaches, which could be said to be a mixed approach. First of all, data collection has been conducted by actual presence in the case company. The data were collected using the check sheet. The team has collected the two months data from the case company.

- **Problem:**

The level of quality expectation of the products within the international customers is high. However, the case company quality control system is not as strong as it is supposed to be. The company is not using statistical quality control tools except check sheets. Because of this, there is a significant amount of rejection of products and has resulted in low economic benefit.

The products with defects do not qualify for the international market. Rather, whenever there is a defect in a product, it is pushed to the local market, which has an impact on the supply chain performance generally.

- **Purpose:**

The case study has shown that the quality of shirts could be improved to the extent of making the defect zero if the statistical quality control tools are applied properly.

- **Result:**

The case study team has shown the methodology and the sequence of the application of the statistical quality control tools. Basically, the systematic application of DMAIC, following with check sheet, process flow diagram, fishbone, Pareto Chart, histogram, control limits, and process capability enhancement could bring significant quality improvement in the case company.

Introduction:

The Seven Basic Tools of Quality are a powerful set of graphical techniques recognized for their effectiveness in resolving quality-related issues. These tools are essential in business, engineering, and manufacturing, where quality is paramount. Quality is pragmatically seen as the excellence or superiority of a product or process. It signifies that a product is fit for its intended purpose and meets or exceeds customer expectations. By utilizing these tools, organizations can achieve and maintain high standards, ensuring that their offerings consistently deliver value and satisfaction to customers.

The garment manufacturing industry is a large and most export-oriented field in developing countries. This manufacturing sector is very important in terms of output, export, and employment creation. Wossi Garment Factory is one of the medium-level companies operating in the global supply chain ecosystem. The company produces export-oriented products; however, the products which do not fulfill the quality requirements of the international market are pushed to the local market.

The purpose of this case study is to reduce or eliminate quality problems in the company's products using statistical quality control tools and hence study the impact of quality control application.

Background and Statement of Problem:

The case company is an export-oriented garment industry producing different varieties of garments. As the group has tried to see a four-month production trend, the maximum share of the products is shirts (37.6%), as shown in Table 1 below. The quality problem in this product will have a high economic impact on the company as it has the highest product share. This is the reason why the paper focuses on the shirt product among the company's products.

S/N	Types of product	Month 1	Month 2	Month 3	Month 4	Total production in each product type	percent
1	Shirt	10218	10842	8242	10478	39780	37.6
2	Gown	5694	5486	4966	5122	21268	20.1
3	Coat	4914	5018	3978	2626	16536	15.6
4	Suit	2548	1794	2314	1742	8398	7.9
5	Trouser	3796	2938	3094	1898	11726	11.1
6	Dress	1170	1898	2652	2262	7982	7.6
Total		28340	27976	25246	24128	105690	

Note: Working days: six days in the week and 26 days in the months.

Table 1: Four months production data of the six products.

Therefore, from the above data, shirt production has the highest production percentage when compared with the other six production types. This is why this case study focuses on shirt production in Wossi Garment Factory. The company does not use statistical quality control tools except for the check sheet. This study focuses on applying other statistical tools in order to improve the quality of specific products.

- **The quality problems (defects) include:**

Sewing defects (skip stitch, broken stitch, side seam uneven, wavy stitch are some of the sewing defects.), Part joining defects, Button attachment defects, Embroidery defects, Finishing defects (Fabric defects, Loose sewing threads, Misaligned button defects, Misaligned buttons holes defects, missing button defects, unfinished buttonhole defects, inappropriate trimming defects). 8.6% of shirt products are pushed to the local market because of quality problems or defects.

Check Sheet and Pareto Chart:

The case company is using only check sheet among the statistical quality control tools. By using check sheet, we have collected two months data from the industry. As our investigation, the major defects in the shirt manufacturing process are sewing defects, Part joining defects, Embroidery defects, finishing defects as shown below on the Table 2

S/n	Defects	Week	Days of a week					
			Mon	Tue	Wed	Thurs	Fri	Sat
1	Sewing defects	1	7	8	8	10	11	12
		2	6	8	7	9	9	11
		3	7	6	8	9	11	11
		4	11	9	12	8	7	7
		5	10	13	15	17	13	10
		6	9	15	11	13	10	9
		7	14	10	12	17	9	8
		8	8	11	12	9	7	12
2	Part joining defects	1	3	4	4	6	6	7
		2	4	3	7	6	5	4
		3	5	5	7	6	5	6
		4	3	5	4	7	6	7
		5	7	5	5	8	11	10
		6	10	7	11	5	6	10
		7	2	7	9	8	10	9
		8	8	11	7	3	10	10
3	Button attachment defects	1	4	3	5	4	3	4
		2	5	6	5	4	3	5
		3	4	3	4	5	3	5
		4	5	4	6	6	5	5
		5	5	7	6	9	10	7
		6	5	7	7	9	7	6
		7	4	7	5	6	9	7
		8	11	8	5	9	6	8
4	Embroidery defects	1	3	2	2	5	4	4
		2	0	2	4	3	1	3
		3	2	3	3	2	2	3
		4	1	3	5	2	4	2
		5	4	3	6	2	4	4
		6	5	6	5	6	4	3
		7	0	3	5	4	1	7
		8	4	4	5	7	6	5
5	Finishing defects	1	4	5	8	9	8	6
		2	5	6	7	6	7	6
		3	6	6	5	7	6	8
		4	6	7	5	6	7	6
		5	9	10	8	9	7	10
		6	7	8	7	10	7	11
		7	7	6	8	11	8	8
		8	9	10	9	11	8	9

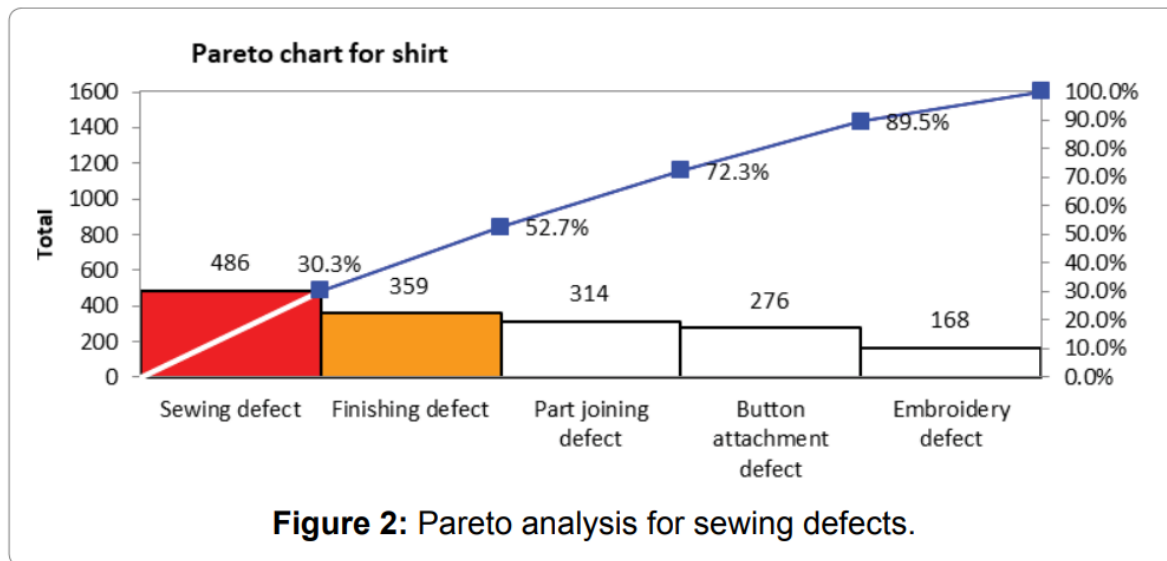
Table 2: Defects collected using check sheet.

As the data on the table above shows, out of the two months' total shirt production, which is 18,720, 1,603 are defective which 8.6% of the total production is. To identify the frequency of occurrence of each defect, we have used Pareto chart (Table 3).

S/n	Defects	Total	Percentage for each	Percentage cumulative
1	Sewing defect	486	30.3	30.3
2	Finishing defect	359	22.4	52.7
3	Part joining defect	314	19.6	72.3
4	Button attachment defect	276	17.2	89.5
5	Embroidery defect	168	10.5	100

Table 3: Shirt defect data for Pareto chart.

The Pareto chart in Figure 2 shows that the most frequent defect is sewing defect, which takes 30.3 % of defects.



Therefore the team decided to analyze sewing defects further by collecting data separately in each sewing defect types. Further data again are collected using the check sheet as

S/n	Defects	Week	Days					
			Mon	Tue	Wed	Thurs	Fri	Sat
1	Skip stitch	1	0	1	0	1	0	1
		2	0	1	2	0	1	0
		3	1	0	1	1	0	0
		4	0	2	1	0	0	2
		5	1	0	1	1	0	1
		6	0	1	1	0	2	0
		7	1	0	1	0	1	0
		8	0	2	1	1	0	1
2	Broken stitch	1	1	0	2	0	0	4
		2	2	3	1	1	2	2
		3	0	1	2	0	3	0
		4	2	1	4	1	3	3
		5	1	2	3	1	2	4
		6	0	5	4	3	0	4
		7	2	0	4	0	2	2
		8	0	4	0	7	0	2
3	Side seam uneven	1	4	3	4	2	3	4
		2	0	2	4	4	3	2
		3	2	4	4	2	3	4
		4	2	4	0	5	2	3
		5	3	4	3	4	6	5
		6	2	3	3	4	3	3
		7	4	4	5	3	6	5
		8	4	4	2	5	6	5
4	Wavy stitch	1	3	4	3	5	6	4
		2	6	5	4	7	5	3
		3	2	3	2	4	5	3
		4	1	3	4	2	3	2
		5	5	6	4	6	5	4
		6	3	5	6	5	4	3
		7	3	3	5	4	1	7
		8	5	4	4	6	7	5

Table 4: Sewing defects.

shown below. According to the data collected, the sewing defect types are Skip stitch, Broken stitch, Side seam uneven and Wavy stitch (Table 4).

After the data is collected on the sewing defects such as Skip stitch, broken stitch, Side seam uneven and Wavy stitch, Pareto chart was done to identify the most frequently occurring sewing defect type as shown Table 5.

S/n	Defects	Total	Percentage for each	Percentage cumulative
1	Wavy stitch	199	41	41
2	Side seam stitch uneven	166	34.1	75.1
3	Broken stitch	90	18.5	93.6
4	Skip stitch	31	6.4	100

Table 5: Data of sewing defects for Pareto chart.

As shown in Figure 3 of Pareto chart, the most frequent sewing defect type is Wavy stitch. However, the most savior defect is the second frequent defect which is Side seam stitch unevenness according to the qualitative data that we gathered from the quality and marketing personnel. Significant Side seam unevenness is serious defect that results on automatic rejection of product.

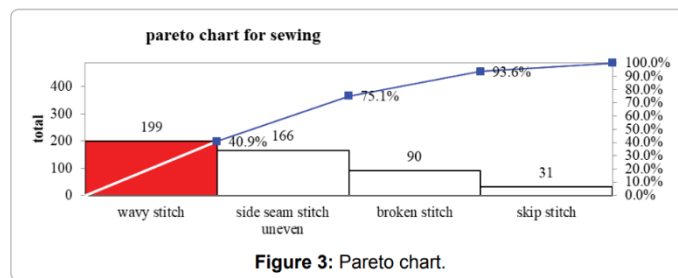


Figure 3: Pareto chart.

Cause and effect diagram:

Both wavy stitch and side seam unevenness need to be solved so that the quality of shirt could be improved. Therefore root cause analysis has been carried out for both wavy stitch and side seam unevenness using fishbone as shown in Figure 4 and the solution is also set for respective root causes (Figure 4).

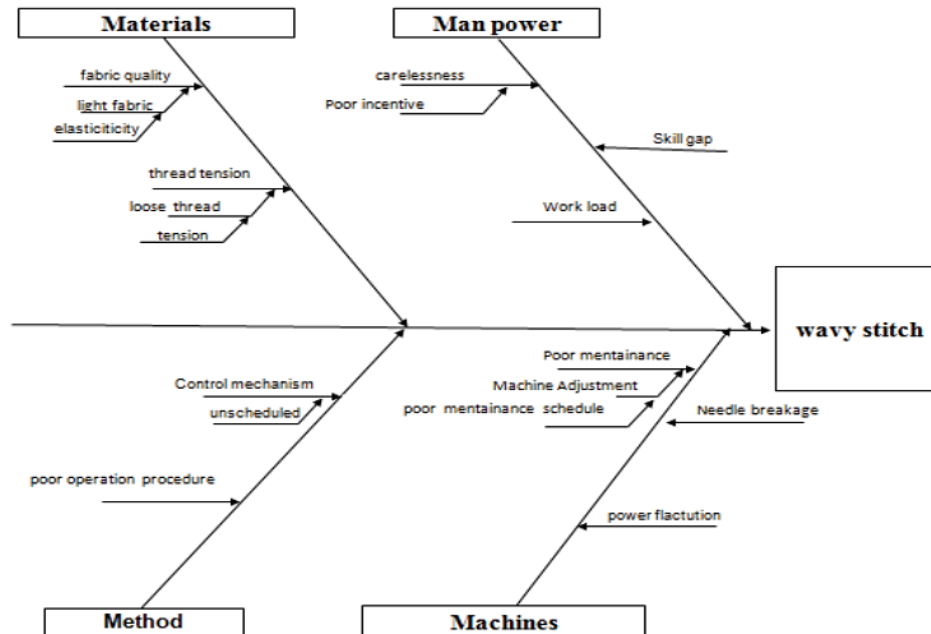


Figure 4: Fishbone for wavy stitch.

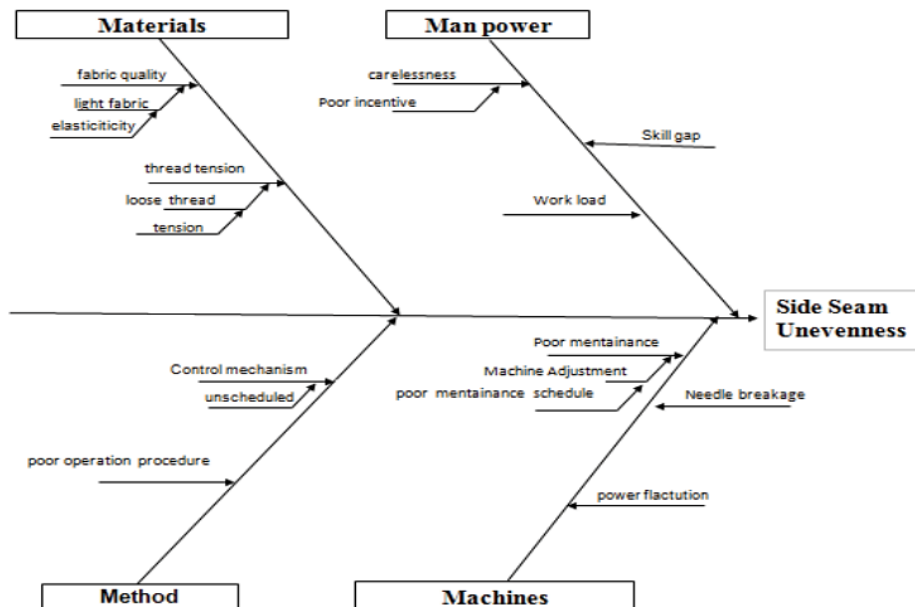


Figure 5: Fishbone for side seam unevenness.

The detail analysis has been done on the root causes and based on the interview with the quality head and the operators, the most savior and frequent root cause of the wavy stitch defect is operator skill gap, carelessness and thread tension. Therefore, we have set the solutions for the root causes as follows: Identify operators with skill gap and providing continuous skills upgrading training, Incentivize and penalize the best performers and careless operators respectively, check the set of the machine daily before straining operation to make sure the tread tension is proper (Figure 5).

The detail analysis has been done on the root causes and based on the interview with the quality head and the operators, the root cause of the side seam unevenness defect is operator skill gap, carelessness and lack of process capability control limit. Therefore, we have set the solutions for the root causes as follows: Identify operators with skill gap and providing continuous skills upgrading training, Incentivize and penalize the best performers and careless operators respectively, determine control limit based on the customer specification and enhance process capability

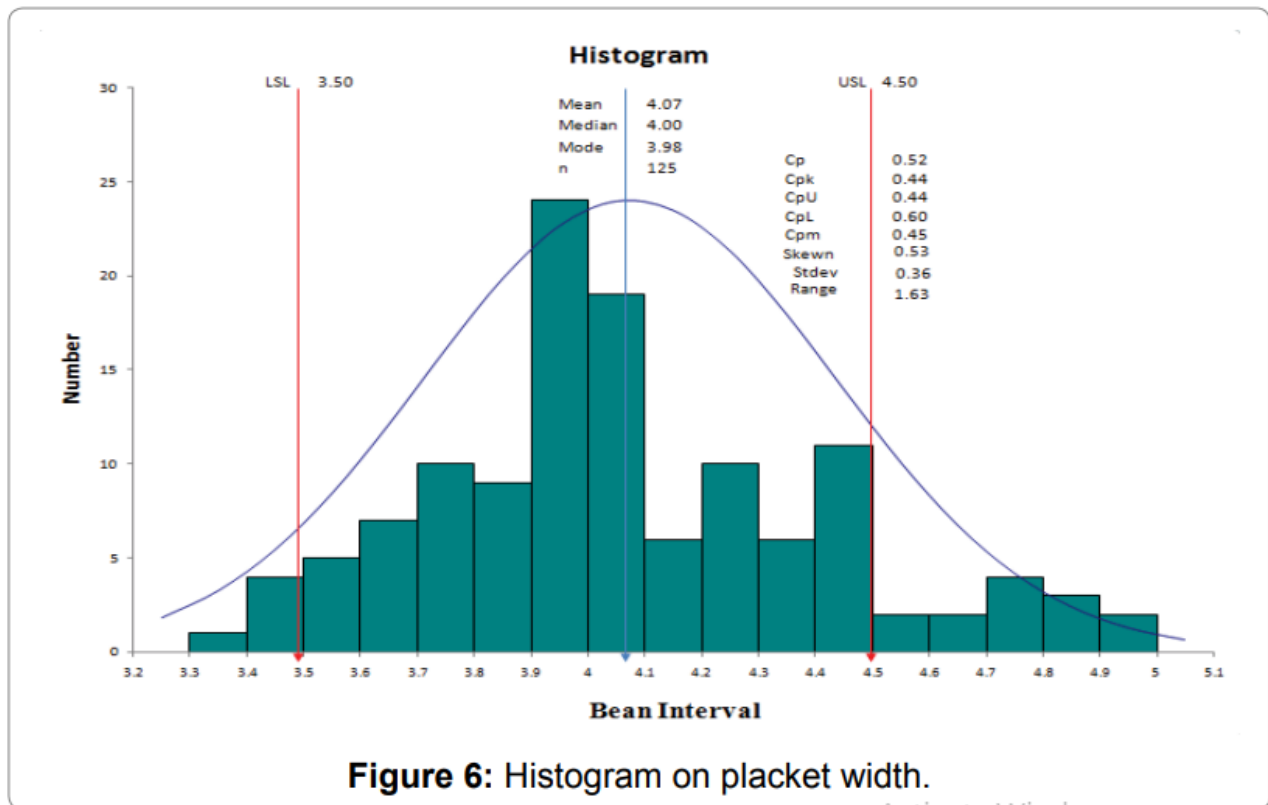
Histogram:

As we are able to understand the most part of the side unevenness that caused for most of product rejection and being out of specification (3.5 - 4.5 cm), the unevenness on placket width is the most serious. Therefore, further data has been collected on the width of the shirt placket to understand the process control and capability. The Table 6 shows that the data collected on the variability of the placket width in centimetre.

Sample strata	1	2	3	4	5	x mean	R
1	3.98	4.11	3.76	4.25	3.6	3.94	0.65
2	4.89	4.5	4.89	3.76	4.5	4.508	1.13
3	4.3	3.5	4.46	3.8	4.3	4.072	0.96
4	3.98	4.1	4.44	3.97	4.23	4.144	0.47
5	3.87	3.98	4	3.87	3.98	3.94	0.13
6	4.23	4.11	3.76	4.45	3.96	4.102	0.69
7	4	3.89	3.75	3.35	3.45	3.688	0.65
8	4.87	4.78	4.35	4.98	4.67	4.73	0.63
9	4	3.87	3.89	4.5	3.67	3.986	0.83
10	3.98	4.2	3.6	4	3.45	3.846	0.75
11	4.5	3.89	3.98	4.02	3.46	3.97	1.04
12	4.45	3.58	4	3.6	4.05	3.936	0.87
13	3.76	4.11	3.76	4.34	4.32	4.058	0.58
14	3.98	4.35	3.76	4.56	3.98	4.126	0.8
15	4.6	3.98	3.87	4.35	3.6	4.08	1
16	4.5	4.03	4.95	4	4.78	4.452	0.95
17	3.98	4.11	3.97	4.25	4.3	4.122	0.33
18	3.99	4.05	3.97	3.67	3.76	3.888	0.38
19	3.96	4.76	3.78	4	3.97	4.094	0.98
20	3.98	4	3.8	4.25	3.5	3.906	0.75
21	3.98	3.5	4	3.98	4.02	3.896	0.52
22	4.4	4.02	3.5	3.98	4.3	4.04	0.9
23	3.98	4	3.69	4.5	3.4	3.914	1.1
24	4.5	3.98	3.76	4	4.7	4.188	0.94
25	3.96	4	4.76	4.25	4.03	4.2	0.8
Total					Mean	4.07304	0.7532

Table 6: Data on variability of width of the placket.

Histogram has been developed using the software and applied to see the distribution of the data. The histogram below is skewed and shows that the distribution of the width of the placket is not normal (Figure 6).



Control chart:

Control chart is developed using the software to check whether the process is statistically in control or out of control. As it is shown Figure 7 on X and R chart, the process was statistically out of control. This shows that there are assignable problems to be solved which are basically investigated using fishbone diagram Figures 4 and 5.

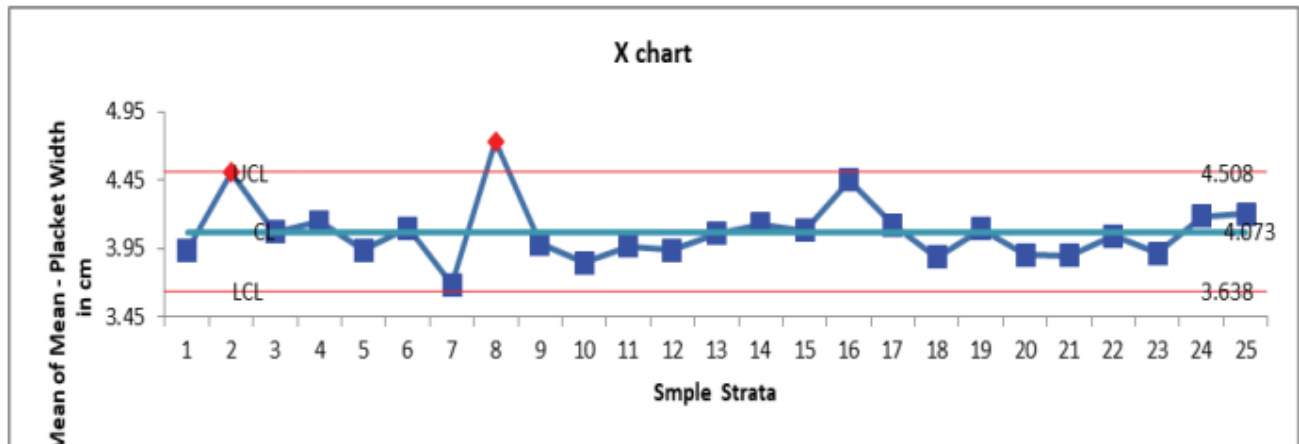


Figure 7: X charts before measurements with assignable problems are not removed.

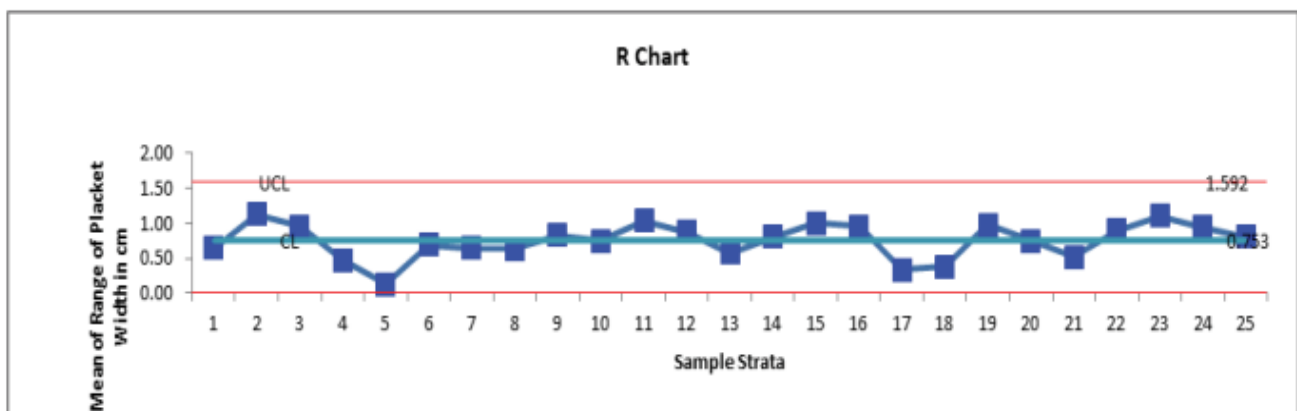


Figure 8: R-charts before measurements with assignable problems are not removed.

Considering that assignable problems will be solved using the solution suggested above following the fish bone analysis technique, we have removed those statistically out of control points and constructed another control limit using the excel add on software as shown below (Figures 8-10 and Table 7).

samples	1	2	3	4	5
1	3.98	4.11	3.76	4.25	3.6
2	4.3	3.5	4.46	3.8	4.3
3	3.98	4.1	4.44	3.97	4.23
4	3.87	3.98	4	3.87	3.98
5	4.23	4.11	3.76	4.45	3.96
6	4	3.89	3.75	3.35	3.45
7	4	3.87	3.89	4.5	3.67
8	3.98	4.2	3.6	4	3.45
9	4.5	3.89	3.98	4.02	3.46
10	4.45	3.58	4	3.6	4.05
11	3.76	4.11	3.76	4.34	4.32
12	3.98	4.35	3.76	4.56	3.98
13	4.6	3.98	3.87	4.35	3.6
14	4.5	4.03	4.95	4	4.78
15	3.98	4.11	3.97	4.25	4.3
16	3.99	4.05	3.97	3.67	3.76
17	3.96	4.76	3.78	4	3.97
18	3.98	4	3.8	4.25	3.5
19	3.98	3.5	4	3.98	4.02
20	4.4	4.02	3.5	3.98	4.3
21	3.98	4	3.69	4.5	3.4
22	4.5	3.98	3.76	4	4.7
23	3.96	4	4.76	4.25	4.03

Table 7: Measurement of placket width after the statistically out of control measurement is removed.

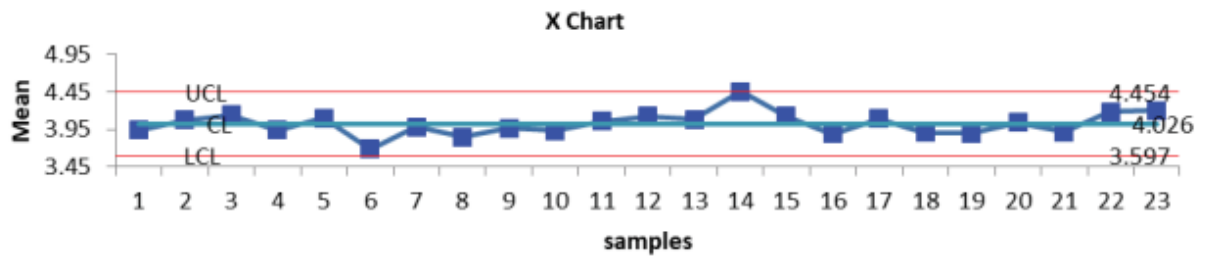


Figure 9: X Chart after the statistically out of control measurement is removed.

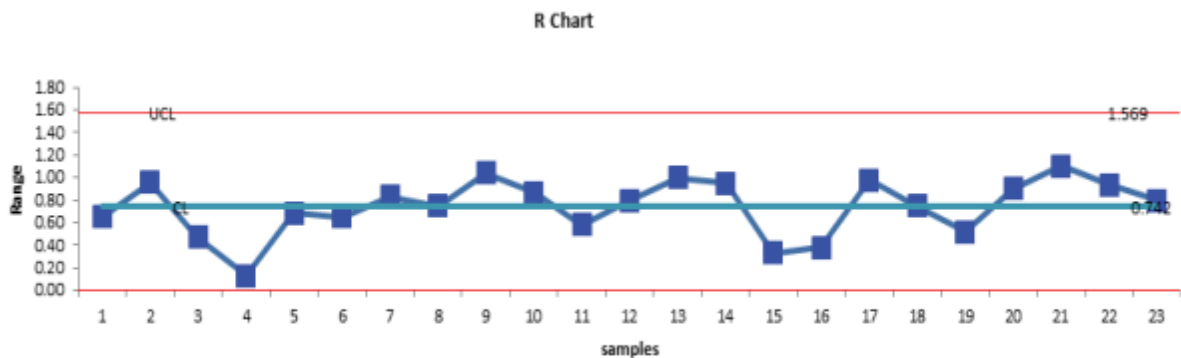
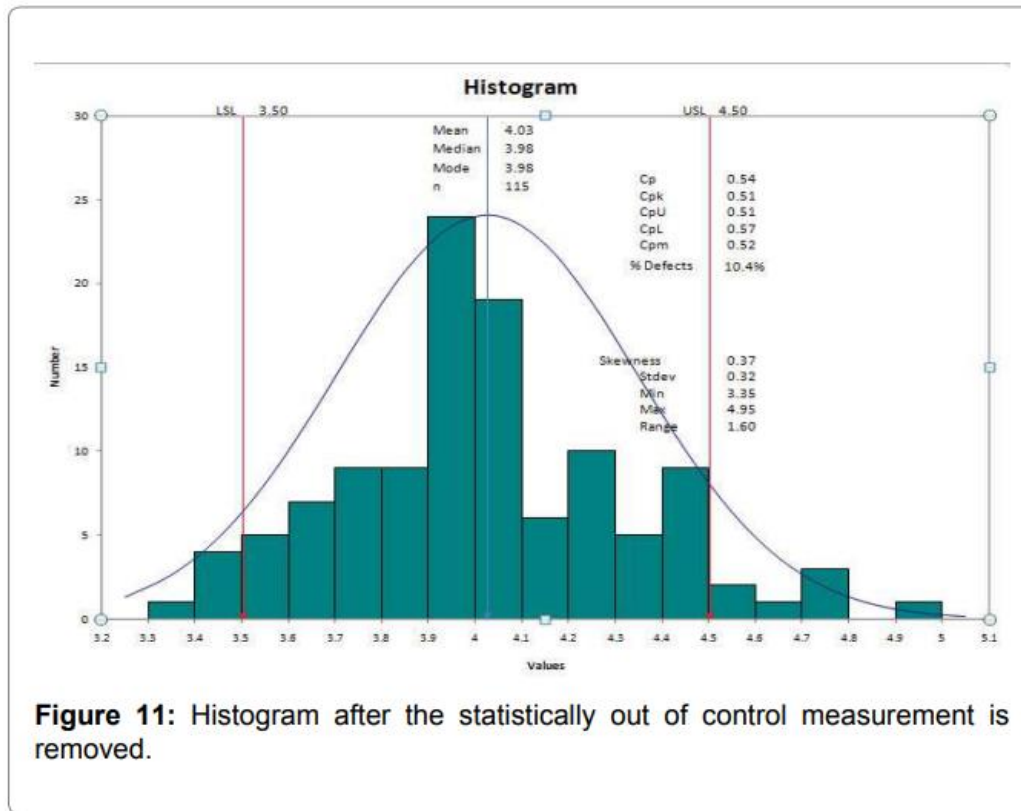


Figure 10: R Chart after the statistically out of control measurement is removed.

The Control charts above show that the data is in statistical control but it does not mean that the process is capable against specification. The specification is 3.5 with LSL and 4.5 with USL where by the target value is 4; all dimensions in cm. Therefore process capability checking has been done using 115 measurements which are in statistically control. Using the data on the table below, Histogram has been developed using the software and the Cp, Cpk, and Cpm analysis has been conducted whether the process is capable or not.

On the histogram Figure 11, we see the value of Cp is greater than Cpk and hence the process is running off center. The histogram is skewed by 0.37. Cp is greater than Cpm and this shows that there are nonconforming units of products and the process is incapable.



By removing those measurements out of specification limits with Consideration that the assignable problems which leads to the process incapability could be solved by the solutions suggested in the fishbone analysis above will be solved, the table below containing 65 measurements has been used to construct final control limit and the process capability has been also checked using the histogram before construction of control limit (Table 8).

The histogram Figure 12 shows that the process is capable as the Cp is less than the Cpm and the value of Cp and Cpk is equal which shows the skewness is almost zero and the process is running at the centre.

Samples	1	2	3	4	5
1	3.98	4.11	3.76	4.25	3.6
2	4.3	3.5	4.46	3.8	4.3
3	3.98	4.1	4.44	3.97	4.23
4	3.87	3.98	4	3.87	3.98
5	4.23	4.11	3.76	4.45	3.96
6	4	3.87	3.89	4.5	3.67
7	4.45	3.58	4	3.6	4.05
8	3.76	4.11	3.76	4.34	4.32
9	3.98	4.11	3.97	4.25	4.3
10	3.99	4.05	3.97	3.67	3.76
11	3.98	4	3.8	4.25	3.5
12	3.98	3.5	4	3.98	4.02
13	4.4	4.02	3.5	3.98	4.3

Table 8: Measurement of Placket width with in specification limit.

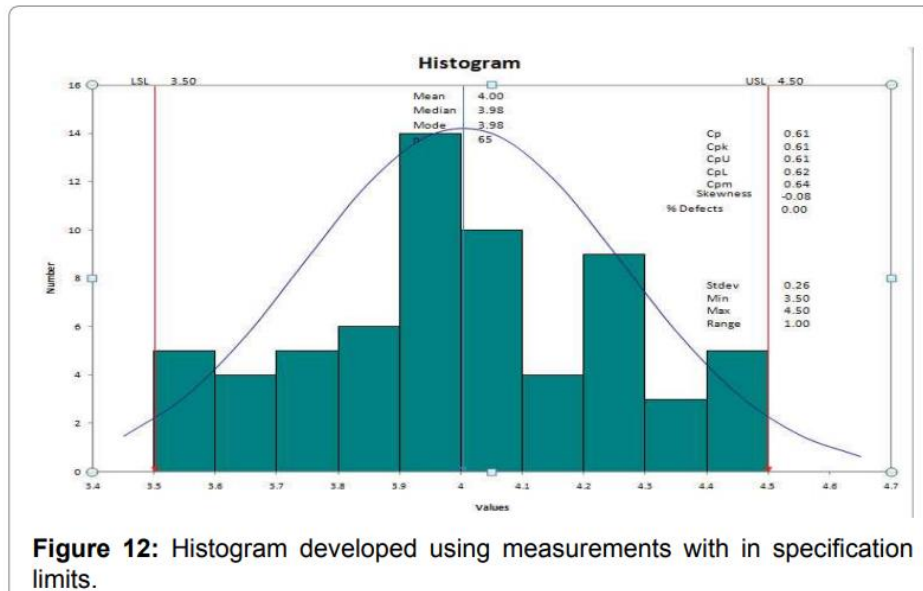


Figure 12: Histogram developed using measurements with in specification limits.

The histogram Figure 12 shows that the process is capable as the Cp is less than the Cpm and the value of Cp and Cpk is equal which shows the skewness is almost zero and the process is running at the centre. Totally 65 measurements with in LSL and USL have been considered to set the control limit for the process capability to be to the required level (Figure 13).

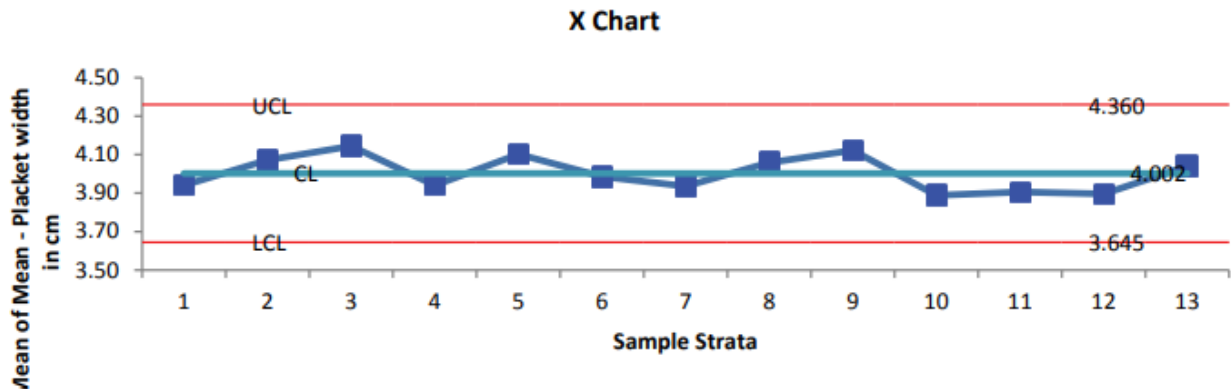


Figure 13: Control Chart after the process capability problems are solved.

Conclusion:

It has been able to see the practices of application of statistical quality control tools and the supply chain management process with perspective of impact of former on the later in the case company. The case company is not following scientific approach of quality control to the level that it is supposed to be although the management is keen about improving the quality. The only statistical quality control tool in use is the check sheet but the rest are not in use. The case study has shown that, the quality of shirt could be improved to the extent of making the defect zero if the statistical quality control tools are applied properly. The case study team has shown the methodology and the sequence of the application of the statistical quality control tools and basically systematic application of DMAIC following with check sheet, process flow diagram, fishbone, Pareto Chart, histogram, control limits and process capability enhancement could bring significant quality improvement in the case company.

Recommendation:

- Based on the conclusion of the case study, the group has forwarded the following recommendations;
- The case company needs to be provide capacity building training for the quality personnel on the statistical quality control tools and their application to improve quality

and also the impact of quality over supply chain improvement and customer satisfaction.

- The case company need to apply SQC tools such as process flow, fishbone, Pareto chart, histogram, control charts in addition to check sheet to improve the quality of product for ultimate satisfaction of customers
- Major root causes of quality problems and reasons for major challenges encountering related to supply chain are workers' skills gap, carelessness and material quality (tread) and hence skills gap training, incentives and quality materials need to be used.